CMPE476 C10K Benchmark Report

1. Test Setup

- Machine: ASUS - System Product Name
- CPU / RAM / Swap: 12 vCPU / 16211 MB RAM / 4096 MB swap
- OS and kernel: Ubuntu on WSL2 (`Linux Duran-Desktop 6.6.87.2-microsoft-standard-WSL2`)
- `ulimit -n`: **65535** (set before the benchmark run)
- Any `sysctl` tuning: none

2. Methodology

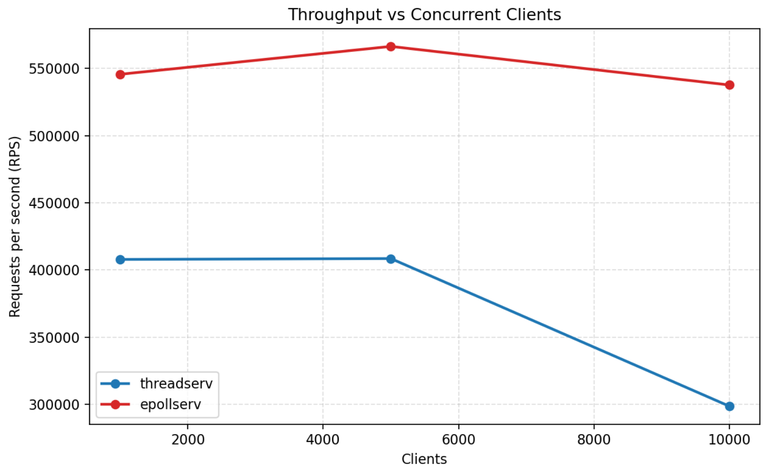
- Build command: `make clean && make && make client_flood`
- Load generator: `client_flood`
- Target concurrency: `1000`, `5000`, `10000`
- Requests per client: `100`
- Host: `127.0.0.1`
- For each server (`threadserv`, `epollserv`) and concurrency level:
  - launch server
  - run `client_flood <host> <port> <clients> <requests_each>`
  - record throughput from `client_flood` output
  - sample `/proc/<pid>/status` during active load and record **peak VmRSS**
  - stop server

3. Results Table

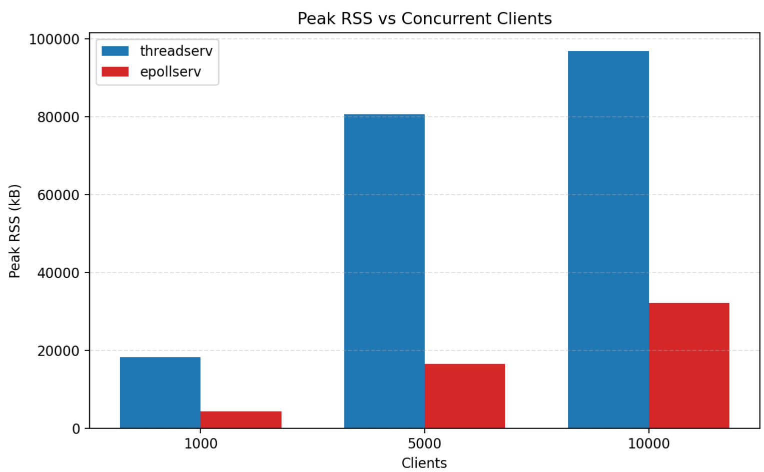
Server	Clients	Requests Each	Comptd. Rqsts.	Elapsed (s)	RPS	RSS Before (kB)	Peak RSS (kB)	RSS After (kB)
t.serv	1000	100	100000	0.245	407862	1408	18304	3904
t.serv	5000	100	500000	1.224	408514	1408	80640	7208
t.serv	10000	100	1000000	3.348	298691	1408	96768	8532
epollserv	1000	100	100000	0.183	545564	1408	4352	3492
epollserv	5000	100	500000	0.883	566296	1408	16512	11632
epollserv	10000	100	1000000	1.860	537594	1408	32128	32128

4. Charts

Throughput vs Clients



Peak RSS vs Clients



5. Analysis

Throughput shows a clear split between the two designs: `threadserv` is roughly flat from 1k to 5k clients (around 408k RPS) but drops noticeably at 10k (around 299k RPS), so thread per connection starts to degrade at high concurrency. `epollserv` stays both faster and steadier (about 546k, 566k, and 538k RPS), which is expected because nonblocking `epoll` avoids creating/scheduling thousands of threads. Memory tells the same story: peak RSS for `threadserv` rises sharply (18,304 -> 80,640 -> 96,768 kB), while `epollserv` stays much lower (4,352 -> 16,512 -> 32,128 kB), so the event-loop model scales better in both throughput and memory at 10k clients.

6. Limitation

The benchmark runs on localhost (WSL2 loopback), so it does not include real network effects such as latency, packet loss, or cross-machine variability; results on a real network may differ.